

JUAN ESTANISLAO HOSZOWSKI

Backend Software Engineer

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SUMMARY

Backend software engineer (Ingeniería en Informática, UBA, Dec 2025) building distributed systems in C++, Python, and TypeScript. Built a RabbitMQ-coordinated pipeline that scaled near-linearly across workers, owned services in a 12-microservice social platform, and wrote a multiplayer game server from scratch with a custom binary protocol.

SKILLS

Languages: C++, Python, TypeScript, Java, Go, SQL

Frameworks: FastAPI, Spring Boot, Express.js, React, React Native

Databases: PostgreSQL, MongoDB, Neo4j, SQLite

Infrastructure: Docker, RabbitMQ, GitHub Actions, CI/CD, Git, Linux, CMake, Render, Firebase

Concepts: Distributed Systems, Microservices, Concurrency and Multithreading, TCP Sockets, Binary Protocol Design, REST APIs, System Design, TDD/BDD, Agile/Scrum

Spoken Languages: Spanish, English, French

PROJECTS

Fault-Tolerant Data Processing System (Team of 3) | [GitHub](#)

Apr 2025 - Jun 2025

Stack: *Python, RabbitMQ, Docker, Pandas, TextBlob*

- Built a distributed pipeline of 40 Docker containers, coordinating workers over RabbitMQ via a custom group-communication middleware to answer 5 analytical queries over 45K movies and 26M ratings.
- Near-linear horizontal scaling: worker counts come from config and the topology is generated from it, so 1→3 workers per stage cut processing time 50% (68s→34s) under 3 clients, no code changes.
- Removed single points of failure with a ring health-check protocol: each node monitors its successor at 1s intervals and restarts crashed containers automatically.

Microservices Social Platform (Team of 5) | [GitHub](#)

Sep 2024 - Dec 2024

Stack: *Python, TypeScript, Java, FastAPI, React Native, PostgreSQL, Neo4j, MongoDB, Docker, Firebase*

- Owned the Twits microservice end to end: 21 REST endpoints covering posts, likes, reposts, comments, feeds, trending, and search, backed by PostgreSQL and Neo4j.
- Shipped 2 of 12 backend services (Twits, Auth) in a polyglot platform spanning TypeScript, Java, and Python, fronted by 3 API gateways and a service registry on Render.
- Architected a polyglot persistence layer: PostgreSQL for relational data, Neo4j for social-graph traversal, MongoDB for real-time chat.

Multiplayer Artillery Game, Worms-style (Team of 4) | [GitHub](#)

Oct 2023 - Dec 2023

Stack: *C++, Box2D, SDL2, Qt, CMake, YAML*

- Owned the networking layer of a 14K-line from-scratch C++ game: a 23-opcode binary TCP protocol with snapshot state sync, and a threaded server running each concurrent match on its own 30 Hz fixed-timestep loop.
- Eliminated client-state divergence under packet loss and latency spikes with authoritative server-side physics (Box2D), which also blocked client-side cheating.
- Shipped 10 weapons with distinct physics behaviors, 18 maps for up to 7 players, and a Qt-based map editor.

EDUCATION

Universidad de Buenos Aires (UBA), Facultad de Ingeniería

Ingeniería en Informática (Computer Engineering)

Buenos Aires, Argentina

March 2018 - Dec 2025